

Dose-dependent effects of decaffeinated coffee on endothelial function in healthy subjects.



BACKGROUND/OBJECTIVES: Coffee is known to contain antioxidant substances whose effects may be blunted because of caffeine that may unfavorably affect the cardiovascular system. This study was designed to investigate the acute dose-dependent effects of decaffeinated coffee (DC) on endothelial function measured by the brachial artery flow-mediated dilation (FMD). **SUBJECTS/METHODS:** A total of 15 (8 men and 7 women) healthy nonobese subjects underwent a single-blind, crossover study. Subjects ingested one and two cups of decaffeinated Italian espresso coffee in random order at 5- to 7-day intervals. **RESULTS:** In the hour following the ingestion of two cups of DC, FMD increased (mean $\pm$ s.e.m.): 0 min, 7.4 $\pm$ 0.7%; 30 min, 8.0 $\pm$ 0.6%; 60 min, 10.8 $\pm$ 0.8%; $P<0.001$) as compared to consumption of one cup of DC (0 min, 6.9 $\pm$ 0.7%; 30 min, 8.4 $\pm$ 1.2%; 60 min, 8.5 $\pm$ 1.1%; 3 x 2 repeated-measures analysis of variance: $P=0.037$ for time x treatment effect). Blood pressure did not differ between groups, and basal heart rate was lower in the two-cup group at baseline and 60 min. **CONCLUSIONS:** The present study demonstrated a significant acute favorable dose-dependent effect of decaffeinated espresso coffee on endothelial function. Further studies are needed to investigate the effects of chronic use of DC especially with respect to caffeinated coffee and in subjects with cardiovascular diseases.